

[ARCHIVED CATALOGUE]

Electrical Engineering (Computer Architecture) (MS)

The Master of Science in Electrical Engineering (Computer Architecture) is earned by successfully completing the normal requirements for the Master of Science in Electrical Engineering, with the following additional requirements: a minimum of 18 units from the Computer Architecture area are required. A breadth requirement of at least one course from Networks and at least one course from VLSI/CAD areas is required. Additionally, with adviser approval, [EE 590](#) Directed Research under the guidance of a Computer Architecture faculty (maximum 4 units) and/or Special Topics [EE 599](#) (maximum 4 units) may also count toward the 18 units required in the Computer Architecture area.

No more than three courses (maximum 12 units) may be counted at the 400 level.

The remaining course work to complete the degree may come from any of the three computer engineering areas.

Computer Architecture

Take at least 18 units from Computer Architecture

- [EE 451 Parallel and Distributed Computation](#) Units: 4
- [EE 454L Introduction to System-on-Chip](#) Units: 4
- [EE 457 Computer Systems Organization](#) Units: 4
- [EE 532 Wireless Internet and Pervasive Computing](#) Units: 3
- [EE 533 Network Processor Design and Programming](#) Units: 4
- [EE 542 Internet and Cloud Computing](#) Units: 4
- [EE 557 Computer Systems Architecture](#) Units: 4
- [EE 554 Cyber-Physical Systems: A Computing Perspective](#) Units: 4
- [EE 560L Digital System Design](#) Units: 4
- [EE 653 Advanced Topics in Microarchitecture](#) Units: 3
- [EE 657 Parallel and Distributed Computing](#) Units: 3
- [EE 659 Advanced Topics in Interconnection Network Design and Analysis](#) Units: 4
- [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2

Networks

Take at least one course from Networks

- [EE 450 Introduction to Computer Networks](#) Units: 4
- [EE 467 Introduction to Communication Systems](#) Units: 4
- [EE 533 Network Processor Design and Programming](#) Units: 4
- [EE 550 Data Networks: Design and Analysis](#) Units: 4
- [EE 555 Broadband Network Architectures](#) Units: 3
- [EE 560L Digital System Design](#) Units: 4
- [EE 597 Wireless Networks](#) Units: 4
- [EE 650 Advanced Topics in Computer Networks](#) Units: 3

VLSI/CAD

Take at least one course from VLSI/CAD

- [EE 477L MOS VLSI Circuit Design](#) Units: 4
 - [EE 536a Analog Integrated Circuits](#) Units: 4
 - [EE 536b Analog Integrated Circuits](#) Units: 4
 - [EE 537 Modern Solid-State Devices](#) Units: 4
 - [EE 552 Asynchronous VLSI Design](#) Units: 4
 - [EE 560L Digital System Design](#) Units: 4
 - [EE 577a VLSI System Design](#) Units: 4
 - [EE 577b VLSI System Design](#) Units: 4
 - [EE 581 Mathematical Foundations for System Design: Modeling, Analysis, and Synthesis](#) Units: 4
 - [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2
 - [EE 680 Computer-Aided Design of Digital Systems](#) Units: 4
 - [EE 580 System Verification](#) Units: 4
 - [EE 582 CMOS:Nano Neuromorphic Circuits](#) Units: 4
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